

MongoDB - Authentication & Authorization - Complete Guide (Roman Urdu)

Part 1: Basic Concept - Authentication vs Authorization

Authentication = Tum kaun ho? (Login check karna). Jaise Facebook me ID password dalna.

Authorization = Tum kya kar sakte ho? (Permission dena). Jaise kisi ki post delete kar sakte ho ya nahi.

Part 2: MongoDB ke Saare Roles (PDF ke hisab se)

General Syntax User Banane ka:

```
db.createUser({
  user: "haroon",
  pwd: "12345",
  roles: [
    { role: "readWrite", db: "school" }
  ]
})
```

1. Database User Roles

read: Sirf data dekh sakta hai. find() kar sakta hai, insert/update/delete nahi kar sakta.

readWrite: Data dekh bhi sakta hai aur change bhi kar sakta hai. Sab se common role hai. Powers: find(), insert(), update(), delete()

```
{ role: "read", db: "school" }
{ role: "readWrite", db: "school" }
```

2. Database Administration Roles

dbAdmin: Database ko manage karta hai, data ko nahi. create indexes, view statistics, collection management kar sakta hai. Lekin user create nahi kar sakta ■

dbOwner: Ek database ka malik. Iske paas 3 roles ka power hai: readWrite + dbAdmin + userAdmin. Data operation + index management + user management.

```
{ role: "dbOwner", db: "school" }
```

3. User Administration Roles

userAdmin: Sirf users ko sambhalta hai - create users, delete users, grant roles. Isko data read/write automatically nahi milta ■

userAdminAnyDatabase: Har database ke users manage kar sakta hai. Hamesha admin database par banta hai.

```
{ role: "userAdmin", db: "school" }
{ role: "userAdminAnyDatabase", db: "admin" }
```

4. Cluster Administration Roles

clusterAdmin: replica sets aur sharded clusters manage

clusterManager: cluster ka configuration manage

clusterMonitor: Sirf monitoring - server status, metrics dekh sakta hai, configuration change nahi kar sakta ■

```
{ role: "clusterAdmin", db: "admin" }
```

```
{ role: "clusterManager", db: "admin" }
{ role: "clusterMonitor", db: "admin" }
```

5. Backup & Restore Roles

backup: Sirf backup lene ke liye

restore: Backup ko wapas restore karne ke liye

```
{ role: "backup", db: "admin" }
{ role: "restore", db: "admin" }
```

6. All Database Roles (Sab Databases ke liye)

readAnyDatabase: Sab databases read kar sakta hai

readWriteAnyDatabase: Sab databases me read/write

dbAdminAnyDatabase: Sab databases ko administer

```
{ role: "readAnyDatabase", db: "admin" }
{ role: "readWriteAnyDatabase", db: "admin" }
{ role: "dbAdminAnyDatabase", db: "admin" }
```

7. Superuser Role - root

root: Sab se powerful. Super Admin. Kuch bhi kar sakta hai. Is role ko kabhi bhi normal user ko mat dena. Sirf apne local PC par use karo.

```
{ role: "root", db: "admin" }
```

Part 3: Practical Setup - Pehle Root Admin Banao, Phir Authorization Enable Karo

Agar pehle authorization ON kar doge to tum khud lock ho jaoge. Isliye sequence ye hai:

STEP 1: MongoDB ko bina Auth ke start karo

Naye install par by default auth OFF hota hai.

STEP 2: Root Admin Banao

```
mongosh

use admin

db.createUser({
  user: "rootAdmin",
  pwd: "Haroon@123",
  roles: [
    { role: "root", db: "admin" }
  ]
})

db.getUsers() // check karne ke liye
exit
```

STEP 3: Authorization Enable Karna (.cfg / .conf File me)

Windows pe: C:\Program Files\MongoDB\Server\7.0\bin\mongod.cfg

Linux pe: /etc/mongod.conf

File ko Admin / sudo me open karo aur ye add karo (space ka dhyan rakho - YAML hai):

```
# mongod.conf / mongod.cfg
storage:
  dbPath: C:\data\db

security:
  authorization: "enabled"
```

STEP 4: MongoDB Restart Karo

```
# Windows
net stop MongoDB
net start MongoDB

# Linux
sudo systemctl restart mongod
```

STEP 5: Ab Login Test Karo

```
# Tarika 1 - Direct login
mongosh -u rootAdmin -p Haroon@123 --authenticationDatabase admin

# Tarika 2 - mongosh ke andar
use admin
db.auth("rootAdmin", "Haroon@123")
// agar 1 return hua to success

# Ab is root se dusre users banao
use school
db.createUser({
  user: "haroon_read",
  pwd: "12345",
  roles: [{ role: "read", db: "school" }]
})
```

Important Notes / Common Galti

1. security ke neech 2 space dena zaruri hai.
2. authorization: enabled ke baad khali mongosh likhoge to 'command not authorized' error aayega - hamesha -u -p ke sath login karo.
3. Practical Tip: School project ke liye readWrite do, admin panel ke liye dbOwner, root sirf local PC par.

End of Guide - Prepared for Haroon Rasheed